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An end-to-end machine learning project for predicting Type 2 diabetes risk using Explainable AI (SHAP & LIME), model evaluation, and an interactive Streamlit dashboard.

[diabetes-risk-prediction-xai-delta.vercel.app](#)

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Diabetes Risk Predictor — Early Detection of Type 2 Diabetes Using Machine Learning and Explainable AI

"Know your risk before it becomes your reality." An end-to-end clinical decision-support system that predicts Type 2 diabetes risk and explains every prediction using Explainable AI.

 Repository CI passing  License MIT  Python 3.10+  XGBoost ML  XAI SHAP | LIME



+ Diabetes Risk Predictor

AI-Powered Risk Assessment. Smarter Health Decisions.

An end-to-end machine learning application that predicts the risk of diabetes using advanced models and explainable AI.

- ML-Powered**
Trained on real-world health data
- Risk Assessment**
Predicts diabetes risk with high accuracy
- Explainable AI**
SHAP insights for transparent predictions
- Fast & Reliable**
Built with FastAPI for performance

Early Prediction. Better Prevention. Healthier Tomorrow.
Empowering individuals and healthcare systems with data-driven insights.

The graphic features a laptop displaying a 'DIABETES RISK' dashboard with a 72% High Risk indicator and 89% Model Confidence. It also shows 'RISK FACTORS' (BMI: 0.72, Age: 0.64, Glucose Level: 0.61, Physical Activity: 0.48, Family History: 0.36) and a 'SHAP SUMMARY' plot. In the foreground, a glucose meter shows a reading of 120 mg/dL next to a test strip and a pen.

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Elevator Pitch

Diabetes is a leading global health crisis, affecting over 500 million people worldwide, yet early detection remains a challenge. This project develops an explainable machine learning system that predicts Type 2 diabetes risk using data from the CDC's Behavioral Risk Factor Surveillance System (BRFSS) 2023. By combining powerful gradient boosting models with SHAP and LIME explanations, the system provides both accurate predictions and transparent, interpretable insights to support clinicians and public health professionals in early intervention.

Business Understanding

Why Diabetes Matters

Type 2 Diabetes Mellitus (T2DM) is one of the fastest-growing chronic diseases globally, with prevalence rising dramatically across all age groups. According to the International Diabetes Federation (2021), over 500 million adults are living with diabetes, and this number is projected to reach 783 million by 2045. Diabetes is a major contributor to cardiovascular disease, kidney failure, stroke, and blindness, placing a significant burden on individuals, healthcare systems, and economies.

Why Prediction Matters

Early identification of individuals at high risk enables targeted preventive interventions, lifestyle modifications, and timely clinical management. However, many cases remain undiagnosed until complications develop. Traditional screening methods often rely on limited clinical criteria and may not effectively identify at-risk individuals across diverse populations. A data-driven predictive model can enhance screening efficiency by leveraging readily available health survey data to prioritize those who would benefit most from further evaluation.

Why Explainability Matters

In healthcare, trust and transparency are paramount. Clinicians and patients need to understand *why* a model makes a particular prediction—not just *what* it predicts. Black-box models, while potentially accurate, can undermine clinical adoption due to a lack of interpretability. Explainable AI (XAI) techniques such as SHAP and LIME address this by providing both global feature importance and local explanations for individual predictions, enabling clinicians to validate model reasoning and make informed decisions.

Why Interpretable AI is Critical for Clinicians

- **Trust:** Clinicians are more likely to adopt a system they can understand and validate.
- **Actionability:** Knowing which features drive a prediction allows for targeted interventions (e.g., "This patient's elevated BMI and hypertension are the primary risk factors").
- **Regulatory Compliance:** Explainability supports auditing and accountability in clinical decision support systems.
- **Patient Communication:** Transparent explanations help clinicians discuss risk factors and preventive strategies with patients.

Data Understanding

Dataset Source

The project uses the **Behavioral Risk Factor Surveillance System (BRFSS) 2023** dataset, a nationally representative health survey conducted annually by the Centers for Disease Control and Prevention (CDC). The dataset captures self-reported information on demographics, health conditions, lifestyle behaviors, and healthcare access from U.S. adults through telephone interviews.

Dataset Details

- **Records:** 433,323 respondents
- **Features:** 350 variables
- **Target Variable:** DIABETE4 (recoded to binary: 1 = diagnosed diabetes, 0 = no diabetes)
- **Selected Features:** 14 predictor variables, including BMI, age, sex, race/ethnicity, general health, physical health, smoking history, physical activity, education, income, hypertension, high cholesterol, kidney disease, and heart disease.

Data Preparation

- Recoded special BRFSS response codes (e.g., "Don't Know", "Refused") to missing.
- Capped BMI values above 60.0 to mitigate extreme outliers.
- Created missingness indicator variables for high-missingness features (income, cholesterol, BMI, smoking).
- Performed median imputation for numerical features and mode imputation for categorical features within a preprocessing pipeline.

Visualizations

Class Distribution

The target variable is imbalanced: ~85.7% non-diabetic, ~14.3% diabetic, reflecting the general population prevalence.

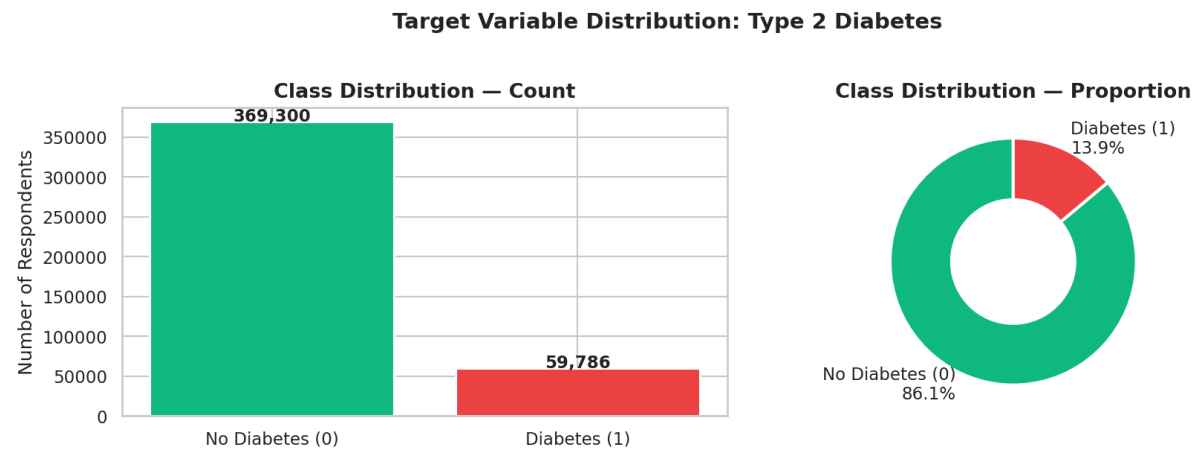


Figure 1. Distribution of the diabetes target variable.

Feature Correlation with Diabetes

To better understand the linear relationships between predictor variables and the target, Pearson correlation coefficients were computed and ranked by magnitude. Hypertension, general health, age, BMI, and high cholesterol demonstrate the strongest positive correlations with diabetes, supporting their inclusion as key predictive features.

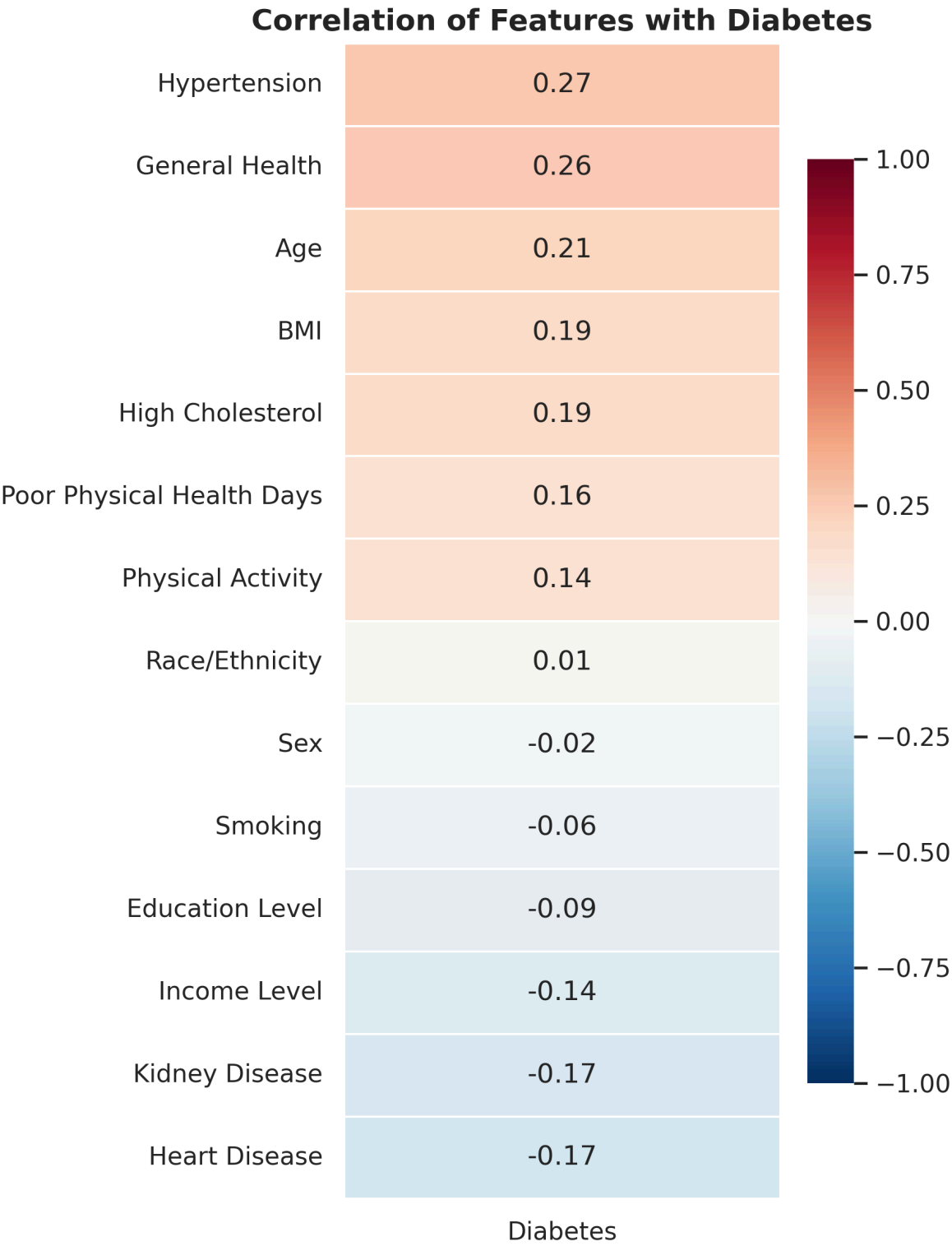


Figure 2. Pearson correlation ranking of predictors with diabetes.

Modeling

Preprocessing Pipeline

A scikit-learn pipeline was constructed to ensure reproducibility and prevent data leakage:

- **Numeric Features:** Standard scaling after median imputation.
- **Categorical Features:** One-hot encoding after mode imputation.
- **Missingness Indicators:** Binary flags for high-missingness features.
- **Class Imbalance:** `class_weight='balanced'` (Logistic Regression, Decision Tree, Random Forest) and `scale_pos_weight` (XGBoost, LightGBM) were applied to address the imbalance.

Model Selection

Four candidate models were trained and compared:

- **Logistic Regression** – Baseline interpretable model.
- **Decision Tree** – Simple non-linear model.
- **Random Forest** – Ensemble of trees with bagging.
- **XGBoost** – Gradient boosting with regularization.
- **LightGBM** – Lightweight gradient boosting.

All models were evaluated on a held-out test set (20% of data).

Hyperparameter Tuning

The best-performing model, **XGBoost**, was further tuned using `GridSearchCV` with 3-fold cross-validation. The optimal hyperparameters were:

- `n_estimators=200`
- `max_depth=4`
- `learning_rate=0.05`
- `subsample=0.85`
- `colsample_bytree=0.85`

Evaluation

Model Performance Comparison

Model	Accuracy	ROC-AUC	Recall	Precision	F1-Score
Logistic Regression	0.729	0.822	0.764	0.315	0.446
Decision Tree	0.680	0.802	0.806	0.283	0.419
Random Forest	0.719	0.824	0.782	0.309	0.443
XGBoost	0.720	0.829	0.797	0.312	0.448
LightGBM	0.716	0.829	0.799	0.309	0.445

Best Model: XGBoost

- **ROC-AUC:** 0.829
- **Recall (Sensitivity):** 0.797 – correctly identifies ~80% of diabetic cases.
- **Precision:** 0.312 – moderate false-positive rate; acceptable for a screening tool where missing a case is more critical.
- **Accuracy:** 0.720

Confusion Matrix

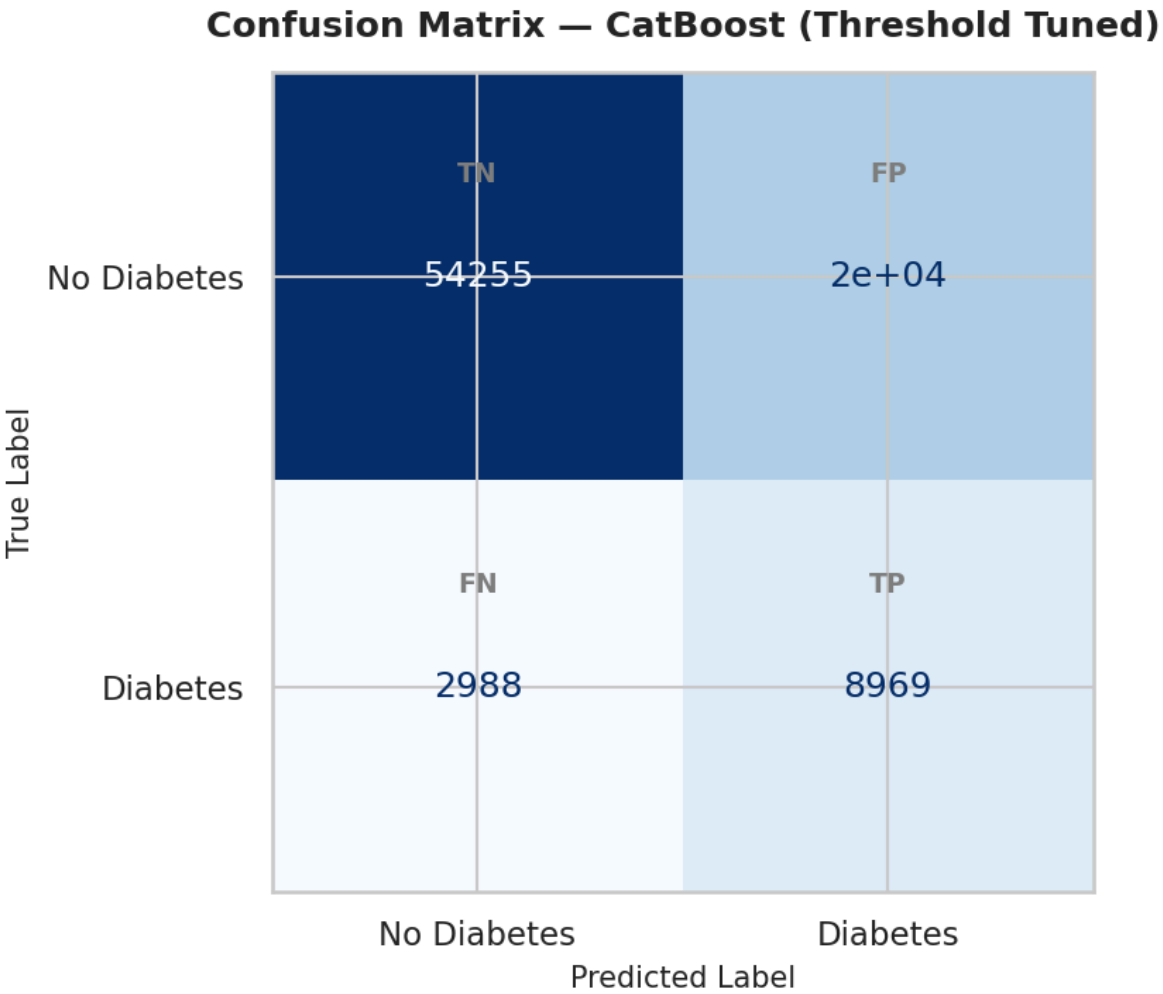


Figure 3. Confusion matrix for the tuned XGBoost model on the held-out test set.

The confusion matrix shows that the model correctly identifies the majority of non-diabetic individuals while maintaining strong sensitivity for diabetic cases, making it suitable as a screening tool.

ROC Curve

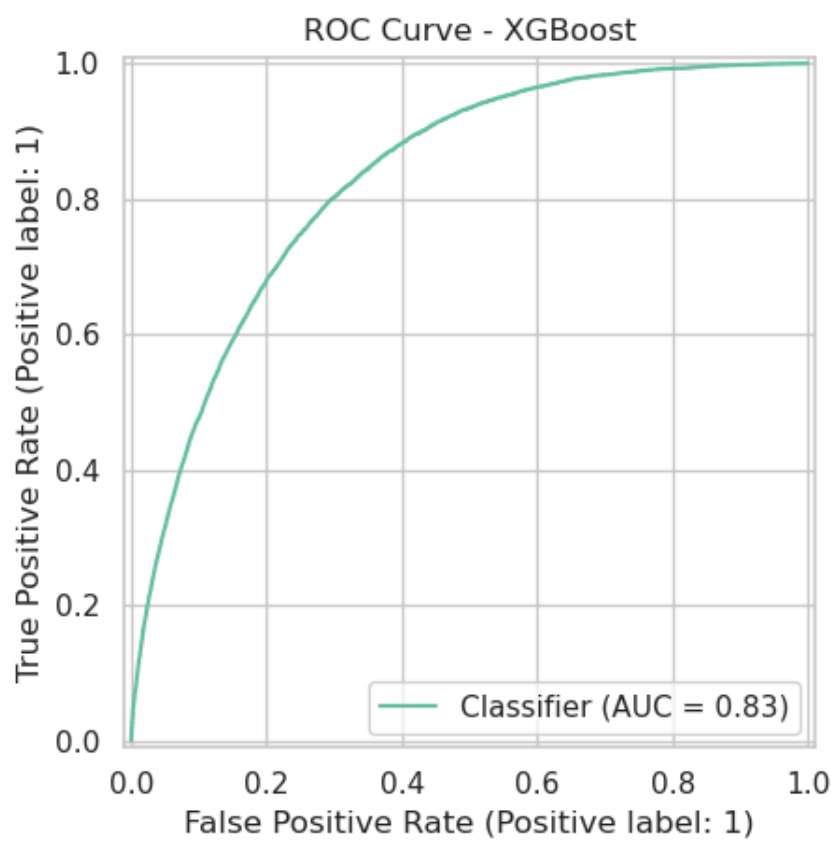


Figure 4. ROC curve for the tuned XGBoost model (AUC = 0.829).

The XGBoost model achieved a ROC-AUC of **0.829**, demonstrating good discrimination between diabetic and non-diabetic individuals.

Model Performance Comparison

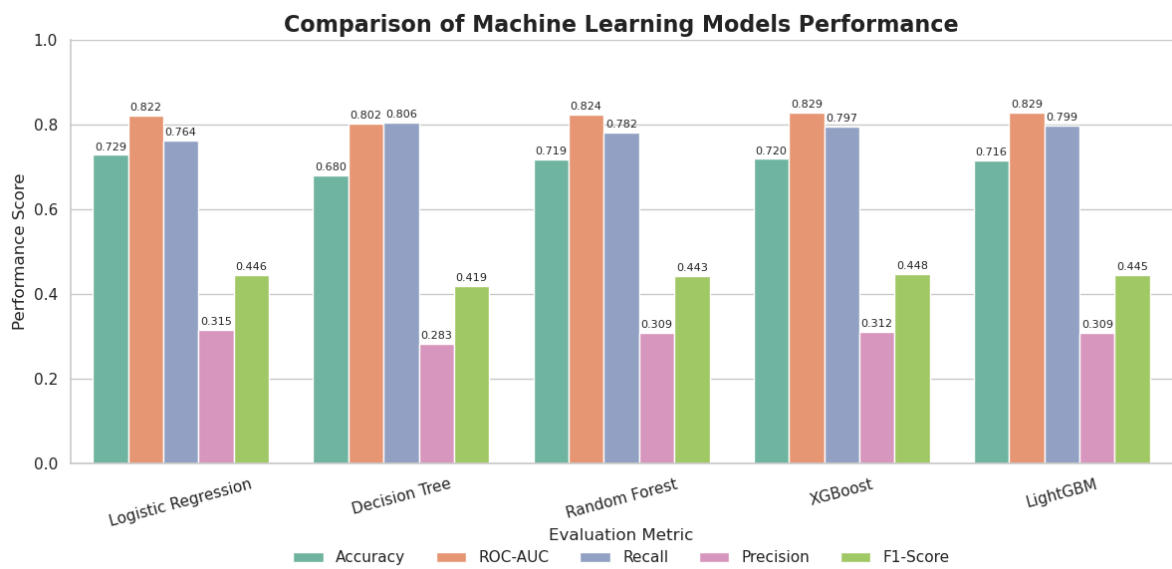


Figure 5. Performance comparison across all candidate models.

XGBoost achieved the highest overall performance, balancing recall and ROC-AUC while maintaining competitive precision and F1-score.

Explainable AI (XAI)

Why XAI Matters

To build trust and facilitate clinical adoption, we employed two complementary XAI techniques:

- **SHAP (SHapley Additive exPlanations):** Provides global and local feature importance based on cooperative game theory.
- **LIME (Local Interpretable Model-agnostic Explanations):** Generates local explanations by approximating the model's behaviour around a specific prediction.

Global Feature Importance (SHAP)

SHAP summary plots reveal that hypertension, general health, age, BMI, and high cholesterol are the most influential predictors across the entire dataset.

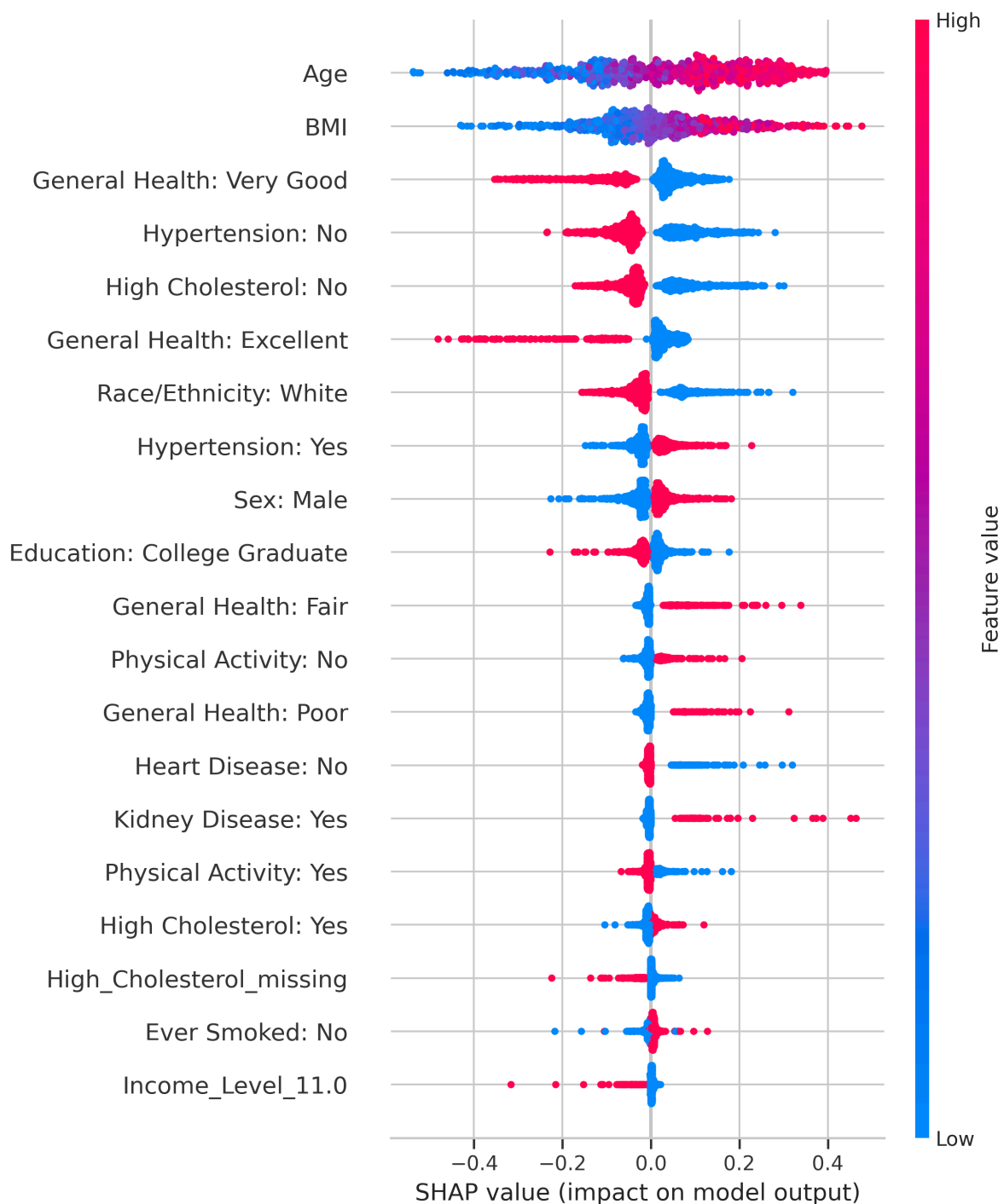


Figure 6. Global SHAP summary plot showing the overall influence and direction of each feature on diabetes risk predictions across the dataset.

The visualization confirms that clinical risk factors such as hypertension, elevated BMI, increasing age, poor general health, and high cholesterol consistently drive model predictions, aligning with established medical knowledge and increasing confidence in the model's interpretability.

Local Explanations

For any individual prediction, SHAP force plots and LIME explanations show which features pushed the prediction toward diabetes or away from it. This allows clinicians to understand the specific risk factors for each patient.

Local Prediction Explanation (SHAP Waterfall)

The SHAP waterfall plot explains how the model arrived at an individual prediction by showing how each feature contributes to moving the prediction from the average (baseline) risk toward the final diabetes risk score. Features pushing the prediction higher are shown in red, while those lowering the predicted risk are shown in blue.

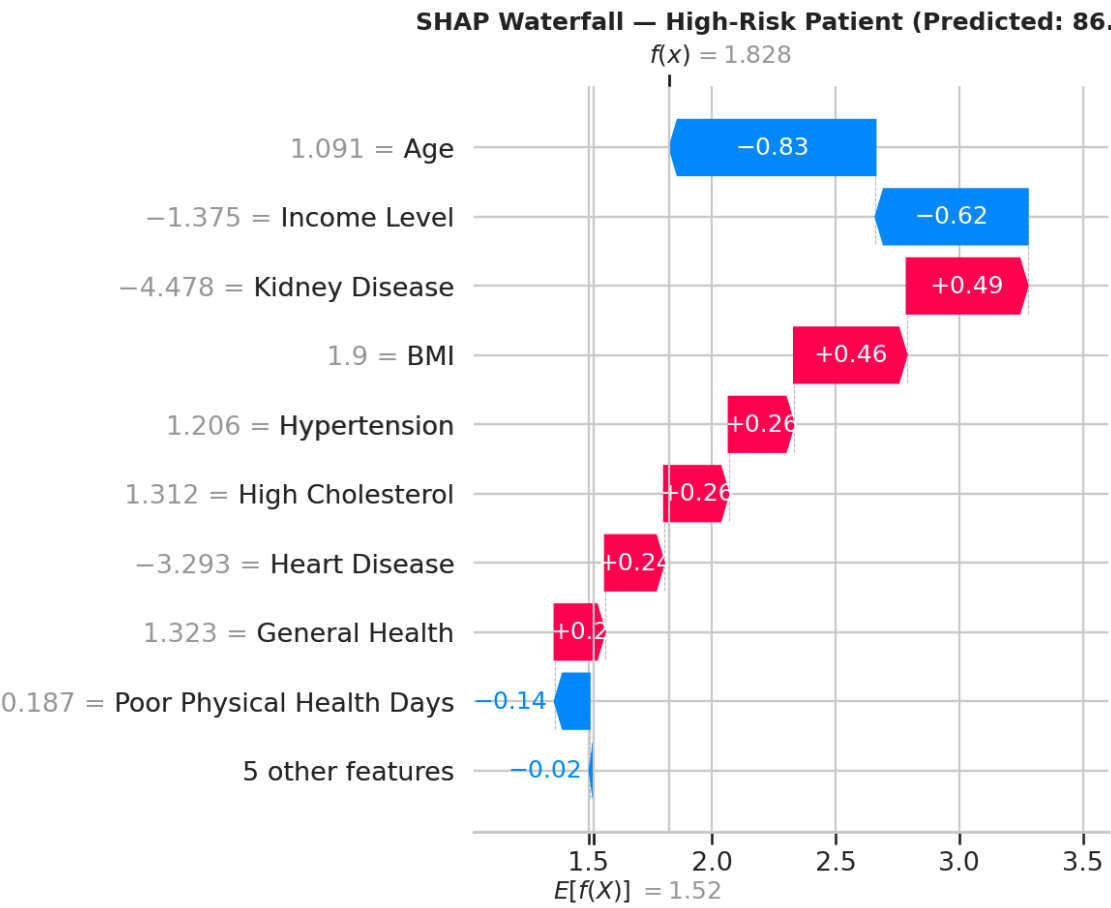


Figure 7. SHAP waterfall plot for a high-risk individual.

For this high-risk individual, hypertension, elevated BMI, increasing age, and poor general health were among the strongest contributors driving the prediction toward diabetes.

Local Prediction Explanation (LIME)

LIME provides a local explanation by approximating the model with an interpretable surrogate model around a single prediction. The visualization highlights the features that most strongly increased or decreased the predicted probability of diabetes for this individual.

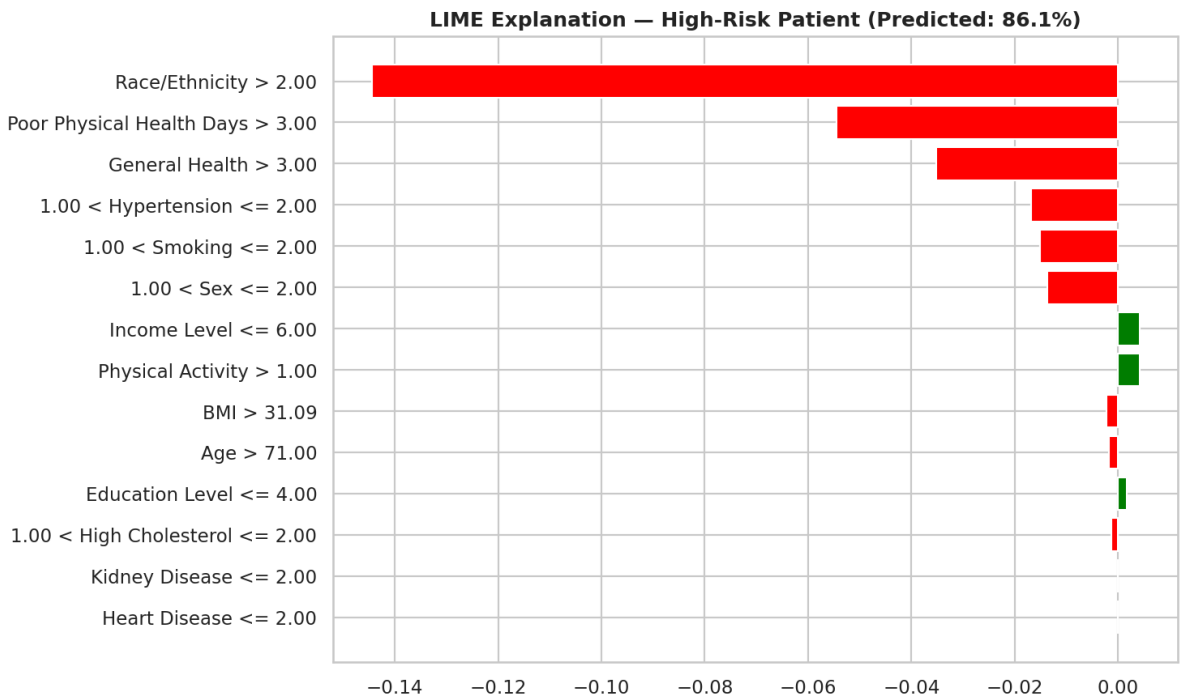


Figure 8. LIME local explanation for the same high-risk individual.

LIME complements SHAP by offering an intuitive explanation of the model's reasoning for an individual patient, enabling clinicians to understand which risk factors most influenced the prediction.

Business Value of XAI

- **Clinical Trust:** Explainable predictions are more likely to be accepted by healthcare providers.
- **Actionable Insights:** Identifying specific drivers (e.g., "high BMI and hypertension") enables targeted interventions.
- **Regulatory Readiness:** Transparent models align with emerging regulations on algorithmic accountability in healthcare.

Conclusion

This project successfully developed an explainable machine learning system for Type 2 diabetes risk prediction using BRFSS data. The XGBoost model achieved a strong ROC-AUC of 0.829 and a recall of 0.797, demonstrating its ability to identify a large proportion of diabetic individuals while maintaining acceptable false-positive rates. The integration of SHAP and LIME provides clear, interpretable explanations at both global and local levels, enabling clinicians and patients to understand and act upon the predictions.

Business Impact: The system can serve as a scalable, evidence-based screening tool to prioritise high-risk individuals for preventive care, reduce undiagnosed cases, and support population health management.

Future Work:

- Deploy the model as a web application for real-time risk assessment.

- Incorporate additional clinical data (e.g., lab results, medical history) for improved accuracy.
- Implement continuous model retraining with new survey data.
- Expand explanations to include patient-friendly language for shared decision-making.

Repository Structure

diabetes-risk-prediction-xai/



```
|
├── .github/                # CI/CD workflows
├── datasets/              # Raw and processed data
├── docs/                  # System design and methodology docs
├── images/                # All visualizations (EDA, modeling, XAI, dashboard)
├── models/                # Trained model, preprocessor, metadata
├── notebooks/             # Final Jupyter notebook and PDF export
├── presentations/         # Slide deck and speaker notes
├── reports/               # Project report and proposal
├── submission/            # PDF exports for submission
├── app/                   # FastAPI backend + Next.js frontend
├── .gitignore              # Excludes data, envs, caches
├── CHANGELOG.md            # Version history
├── CONTRIBUTING.md         # Contribution guide
├── LICENSE                 # MIT License
├── README.md               # Project home page
├── environment.yml         # Conda environment
└── requirements.txt        # Pip dependencies
```

Repository Navigation

-  [Final Notebook](#)
-  [Presentation](#)
-  [Live Application](#)
-  [Backend API](#) (*FastAPI*)
-  [Frontend Dashboard](#) (*Next.js*)
-  [Final Report](#)
-  [Proposal](#)
-  [Documentation](#)

Installation & Reproduction

Prerequisites

- Python 3.8+
- Git

- **Node.js 16+** (Optional: for frontend dashboard)
- **Conda** (Optional: for environment management)

Clone the Repository

```
git clone https://github.com/your-username/diabetes-risk-prediction-xai.git
cd diabetes-risk-prediction-xai
```



Set Up Python Environment

Using Conda:

```
conda env create -f environment.yml
conda activate diabetes-xai
```



Using pip:

```
python -m venv venv
source venv/bin/activate      # On Windows: venv\Scripts\activate
pip install -r requirements.txt
```



Run the Jupyter Notebook

Launch Jupyter Lab or Notebook and open `notebooks/diabetes_prediction.ipynb` to explore the complete analysis, data preparation (mean imputation), SMOTE resampling, pipeline hyperparameter tuning, and XAI explanations.

Run the Backend API (FastAPI)

```
cd app/api
uvicorn main:app --reload
```



The API will be available at `http://localhost:8000` . Interactive API documentation and custom threshold prediction endpoints can be tested at `http://localhost:8000/docs` .

Run the Frontend (Next.js)

```
cd app/web
yarn install
yarn dev
```



The dashboard will be available at `http://localhost:3000` .

Deployment

- **Frontend:** Deployed on Vercel (or Netlify).
- **Backend API:** Deployed on Render, Heroku, or AWS.
- **CI/CD:** GitHub Actions (see `.github/workflows/ci.yml`) runs tests, formatting, and linting on push.

Model Results & Application

This section showcases the deployed application together with the key visualizations that demonstrate the model's performance and interpretability.

Interactive Dashboard

The web application enables users to assess diabetes risk in real time while providing explainable AI insights for each prediction.

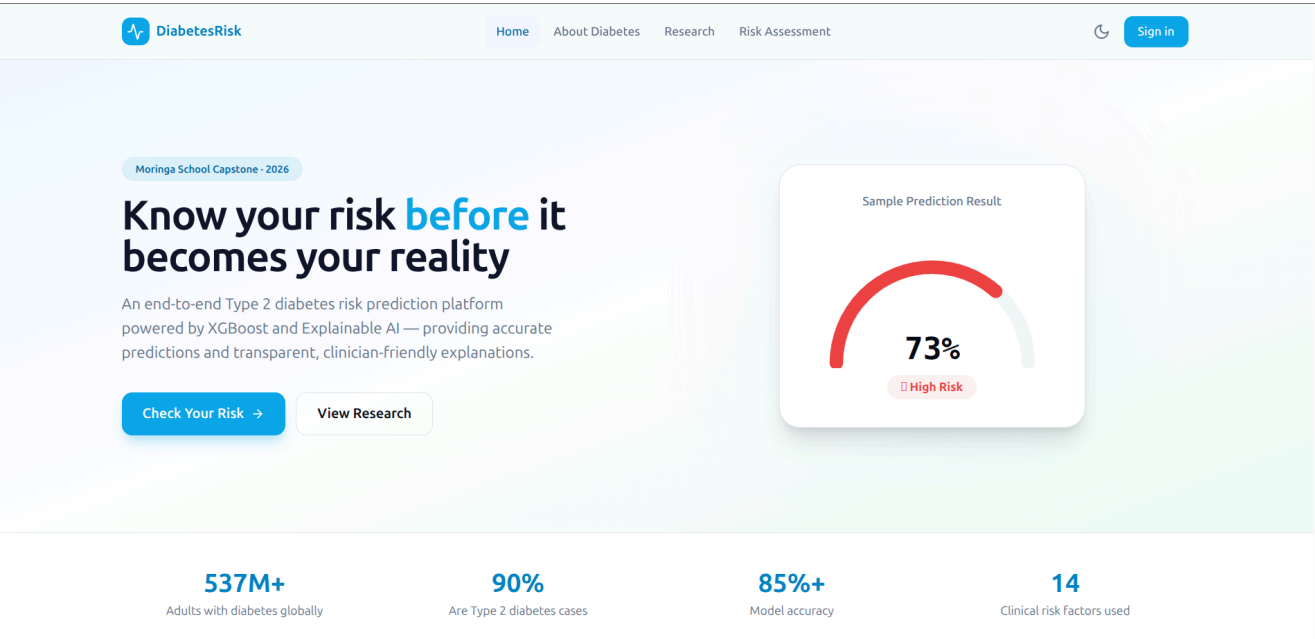


Figure 9. Interactive risk-assessment dashboard.

SHAP Global Feature Importance

The SHAP summary plot highlights the features that have the greatest overall influence on diabetes risk predictions across the entire dataset.

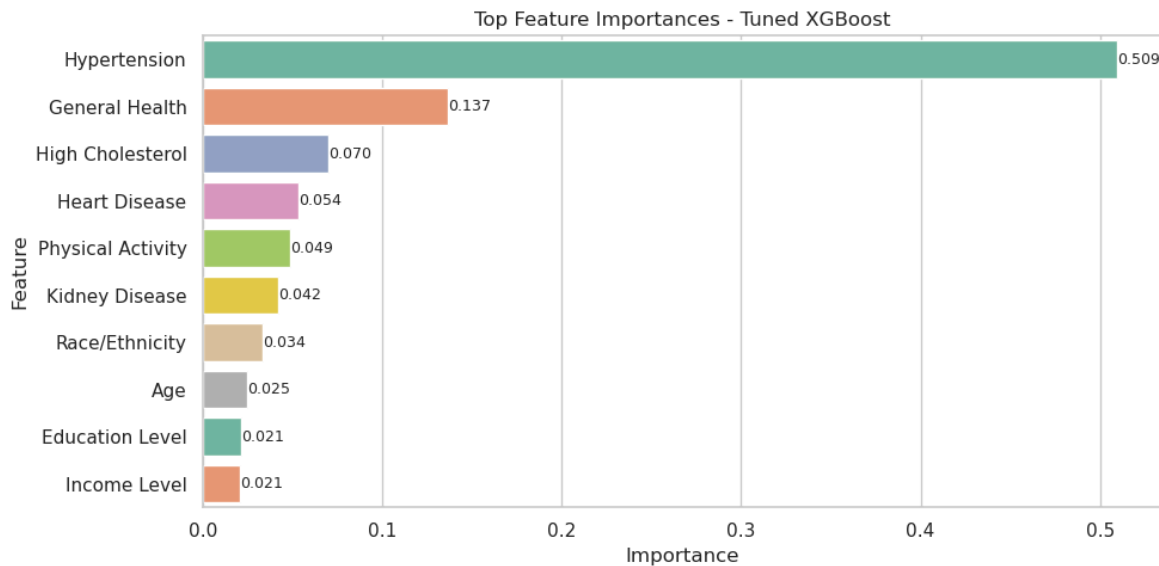


Figure 10. Ranked global feature importance from the tuned XGBoost model.

ROC Curve

The Receiver Operating Characteristic (ROC) curve demonstrates the model's ability to distinguish between diabetic and non-diabetic individuals. The XGBoost model achieved an ROC-AUC of **0.829**, indicating strong discriminative performance.

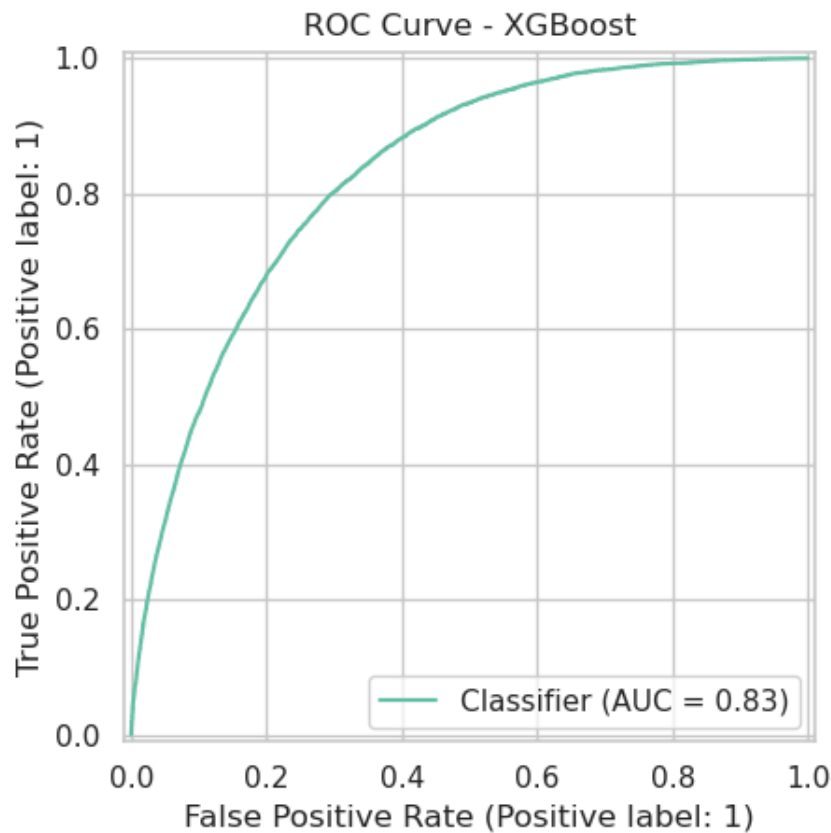


Figure 4 (repeated). ROC curve for the tuned XGBoost model.

🔧 Technologies Used

Machine Learning & XAI

- **XGBoost, LightGBM** – Gradient boosting models optimized via cross-validation pipelines.
- **Imbalanced-Learn** – SMOTE resampling contained within training folds to avoid data leakage.
- **Scikit-learn** – Preprocessing pipelines, data preparation (imputation/scaling), and evaluation metrics.
- **SHAP, LIME** – Model explainability and feature importance analysis.

Data Processing & Analysis

- **Pandas, NumPy** – Data manipulation and structural preparation.
- **Matplotlib, Seaborn** – Exploratory data analysis (EDA) plots and distribution metrics.

Backend

- **FastAPI** – API framework managing continuous outputs and decision thresholds.
- **SQLAlchemy** – ORM (*if database integration*).
- **PostgreSQL** – Database (*if used*).

Frontend

- **Next.js** – React framework for the interactive clinician interface.
- **TailwindCSS** – Clean UI styling.
- **Chart.js / Recharts** – Interactive probability and risk tracking visualizations.

Deployment & DevOps

- **Render** – Backend hosting.
- **Vercel** – Frontend hosting.
- **GitHub Actions** – Continuous Integration (CI) tracking.

📅 Future Work

Releases

No releases published

[Create a new release](#)

Deployments 45

- ✓

Production yesterday
- ✓

Preview 2 days ago

+ 43 deployments

Packages

No packages published

[Publish your first package](#)

Contributors 2

- 

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Languages

- Jupyter Notebook 80.4%
- HTML 15.4%
- TypeScript 2.9%
- Python 1.2%
- JavaScript 0.1%
- Makefile 0.0%